Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

# Results

01 · POISSON / NEGATIVE BINOMIAL

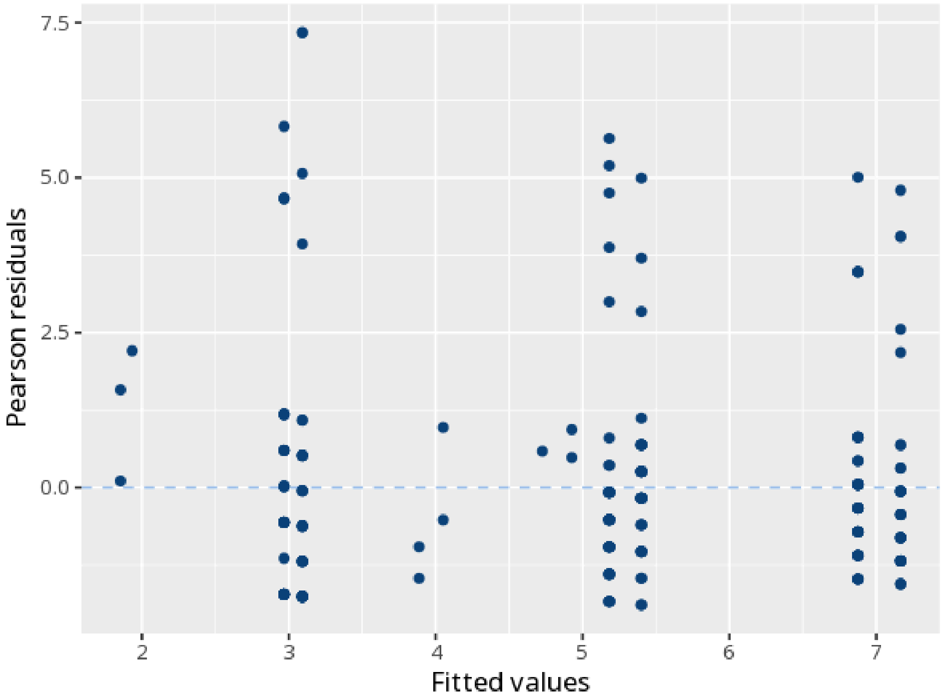
predict a count outcome

**Table 1.** Coefficients

| | Log-count (B) | IRR |
|---|---|---|
| (Intercept) | −1.09 | 0.34 |
| | (0.06) | [0.30, 0.38] |
| teaching_method: Flipped | −0.56 | 0.57 |
| | (0.08) | [0.49, 0.67] |
| teaching_method: Lecture | 0.28 | 1.33 |
| | (0.06) | [1.17, 1.51] |
| gender: Male | −0.04 | 0.96 |
| | (0.06) | [0.86, 1.07] |
| Num.Obs. | 240 | |
| Dispersion | 3.14 | |
| Residual deviance | 633.11 | |
| df | 236 | |
| Log.Lik. | −683.51 | |
| AIC | 1375.03 | |
| BIC | 1388.95 | |

dispersion check (Poisson): if variance >> mean (over-dispersion), switch to negative binomial, which instead reports theta (its overdispersion parameter) rather than this ratio. Overdispersion test: $\chi^2(236)$ test $p < .001$ - overdispersion detected.

**Figure.** Fit / residuals




### Understanding the numbers

**Incidence-rate ratio (IRR).** The multiplicative change in the expected count per one-unit increase in the predictor (>1 more, <1 fewer). Here, IRR = 0.57.

**Log-count (B).** The raw coefficient on the log-count scale that the IRR is exponentiated from. Here, B = −0.56 on the log-count scale.

**Dispersion.** Checks whether the counts are over-dispersed (variance much greater than the mean) - if so, a negative-binomial model (reporting θ instead) fits better than Poisson. Here, dispersion = 3.14, overdispersion test p < .001.

**Residual deviance.** How much variation in the counts remains unexplained by the model - compare it to its degrees of freedom to gauge fit. Here, residual deviance = 633.11.

**df.** The degrees of freedom for the residual deviance, i.e. observations minus estimated parameters. Here, df = 236.

**Log.Lik..** The log-likelihood: how probable the observed counts are under the fitted model; it underlies AIC and BIC. Here, Log.Lik. = −683.51.

**AIC.** A fit measure that penalizes model complexity, useful for comparing models fit to the same data (lower is better). Here, AIC = 1375.03.

**BIC.** Like AIC but penalizes complexity more heavily as sample size grows (lower is better). Here, BIC = 1388.95.

**N.** The number of observations the model was fit on. Here, N = 240.

### How to read this test

Models event counts. Each predictor's incidence-rate ratio (IRR) gives the multiplicative change in the expected count per unit (>1 more, <1 fewer); p tests significance. Check dispersion to pick Poisson vs. negative binomial. If cases have unequal exposure (different observation time/area/population), add an offset of log(exposure) so the model predicts rates, not raw counts - omitting it biases the IRRs. Your significance threshold (α) is 0.05.

**APA template:** Predictor teaching_method: Flipped was associated with the count, IRR=0.57, 95% CI [0.49, 0.67], p < .001.

**Statistical basis:** Poisson regression (incidence-rate ratios) (R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.); Negative binomial regression (glm.nb) for overdispersed counts (Venables WN, Ripley BD (2002). Modern Applied Statistics with S, Fourth edition. Springer, New York.); Overdispersion check (Lüdecke D, Ben-Shachar MS, Patil I, Waggoner P, Makowski D (2021). "performance: An R Package for Assessment, Comparison and Testing of Statistical Models." Journal of Open Source Software, 6(60), 3139.). Full references in the exported CITATIONS.txt.

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